Hardware for Deep Learning

Systems, Chips, Algorithms

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Zurich Office

1. Computing architecture, cover the almost all aspects of computing

Von Neumann Lab

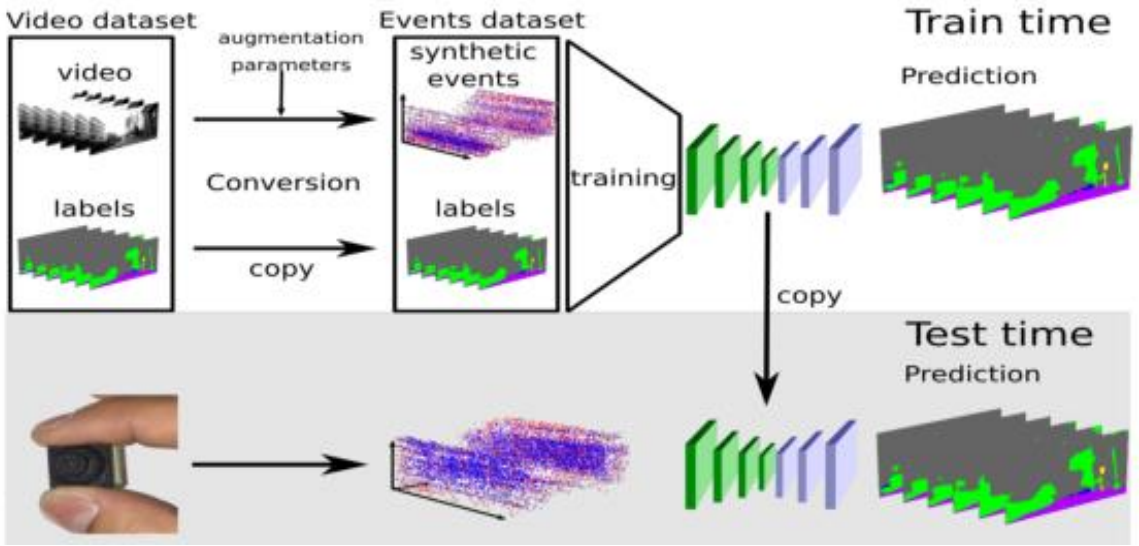
- **Tools and Methodology**
  - Hardware design automation
  - Design Space Exploration
- **Compiler and Software and hardware co-design**
  - Parallelism exploration and DSA mapping
  - Hardware Software Co-Design
- **System structure (HPC)**
  - Performance analysis and improvement
- **Instruction Set**
  - ISA definition and verification(Formal)
- **Microarchitecture**
  - CPU/GPU micro-architecture study for performance and power efficiency
- **Hardware (chip) security**
  - Security architecture study
  - Security beyond Isolation
- **Computing Theory**
  - Basic computing theory study
  - New computing model study

1. Including processor micro-architecture, SoC chip architecture, modeling simulation and verification, chip agile development (including EDA), open source ecology, compiler and tool chain and optimization, system software, security and other computer architecture full-scale cooperation. Short-term key breakthroughs; long-term targeted institutional research.

2. Talent building and training (PhD, postdoc, academic vacation), including Dr. Funding, postdoc. Post-doctoral, scientists and researchers.

Hiring Interns at PhD and Master level:
https://www.linkedin.com/jobs/search/?f_E=1&keywords=huawei

2. Computing Vision:
New algorithm for next generation Tech with partner.



HUAWEI

Outline

1) Quantization

2) Huawei DaVinici Core and Ascend SoC

Number Representations

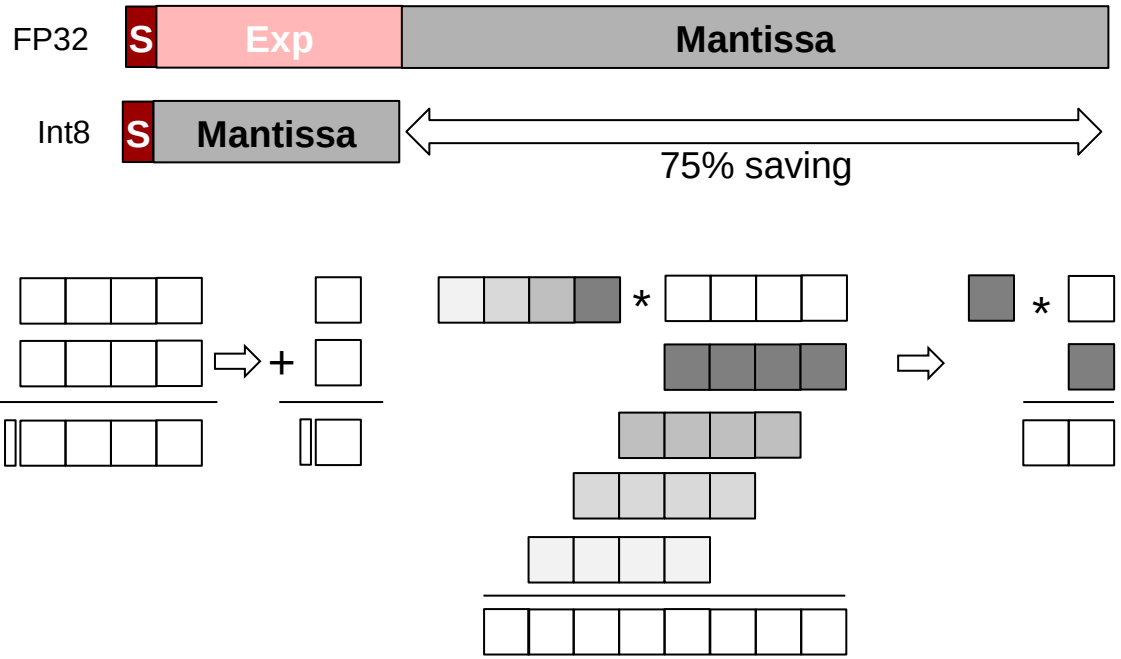
			Range	Worst-case Accuracy
FP32	1 S, 8 Exp, 23 Mantissa	Single-precision Floating-Point	3×10^{-38} ... 3×10^{38}	0.000006%
FP16	1 S, 5 Exp, 10 Mantissa	Half-precision Floating-Point	6×10^{-5} ... 6×10^{5}	0.05%
Bfloat16	1 S, 8 Exp, 7 Mantissa	Brain Float	3×10^{-38} ... 3×10^{38}	0.4%
TF32	1 S, 8 Exp, 10 Mantissa	Tensor Float	3×10^{-38} ... 3×10^{38}	0.05%
Int32	1 S, 31 Mantissa	Signed 32-bit Integer	0 ... 2×10^{9}	33%
Int16	1 S, 15 Mantissa	Signed 16-bit Integer	0 ... 3×10^{4}	33%
Int8	1 S, 7 Mantissa	Signed 8-bit Integer	0 ... 127	33%
Int4	1 S, 3 M	Signed 4-bit Integer	0 ... 7	33%

$$\max rel\ error_{INT} = \frac{|Round(1.5-\varepsilon)-(1.5-\varepsilon)|}{1.5-\varepsilon} = \frac{|1-1.5+\varepsilon|}{1.5-\varepsilon} = \frac{0.5+\varepsilon}{1.5-\varepsilon} \rightarrow \frac{1}{3}$$

$$\max rel\ error_{FP} = \frac{|Round(1+2^{-M-1}-\varepsilon)-(1+2^{-M-1}-\varepsilon)|}{(1+2^{-M-1}-\varepsilon)} = \frac{|1+2^{-M}-1-2^{-M-1}+\varepsilon|}{(1+2^{-M-1}-\varepsilon)} = \frac{2^{-M-1}+\varepsilon}{1+2^{-M-1}+\varepsilon} \approx 2^{-M-1}$$

Why Reduced-Precision Operations?

- **Space savings**
  - Memory (weights + activations/feature maps)
  - Occupies fewer bits on interconnect (memory-compute, compute-compute)
- **Area savings due to simpler arithmetic**
  - Addition of two n-bit integers: O(n)
  - Multiplication of two n-bit integers: $O(n^2)$ (*)
- **Energy savings**
  - Fewer bits need to be moved
  - Energy savings for a given pJ/Bit



	2010 (40nm)	2017 (10nm)
FP64 Fused Multiply-Add	50 pJ	8.7 pJ
64-bit read from 8kB SRAM	14 pJ	2.4 pJ
Wire energy per transition	240 fJ/bit/mm	150 fJ/bit/mm
Wire energy (256 bit, 10 mm, transition prob. 0.5)	310 pJ	150 pJ

6x 17x

Table from [1]

*) School-book style multiplication

[1] Keckler, Dally, et al. GPUs and the Future of Parallel Computing. In IEEE Micro, 31(5), 2010.

Reduced Precision? Where?

- **Weights**
  - Fit more weights on the device (phone, micro-controller)
- **Activations**
  - Feature Maps in CNNs often too large to fit into on-chip memory
  - Example:

    ResNet-50 on [256, 3, 224, 224] input (ImageNet with batch size 256, NCHW)

    Largest Conv2d feature map: [256, 256, 56, 56] x FP32 → 784 MiB!
- **Gradients**
  - Reduce communication cost in **`all-reduce`** operation of data parallel training.
  - 1-bit SGD [2] inspired by $\Sigma\Delta$-modulation, QSDG [3] randomized rounding + Elias coding, TernGrad [4] generates gradient vector with elements $\in\{-1, 0, +1\}$

[2] Seide et al. 1-Bit Stochastic Gradient Descent and its Application to Data-Parallel Training of Speech DNNs. In Interspeech'14.
[3] Alistarh et al. QSGD: Communication-Efficient SGD via Gradient Quantization and Encoding. In NIPS'17.
[4] Wen et al. TernGrad: Ternary Gradients to Reduce Communication in Distributed Deep Learning. In NIPS'17.

Reduced Precision? How?

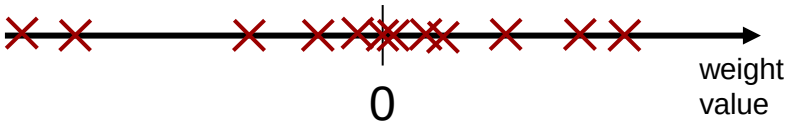
- **FP32 → FP16/Bfloat16/TF32**
  - **Post-training**: just convert trained model (e.g., via TensorFlow Lite `TFLiteConverter`), that usually works directly
  - Training in TF16 or **mix-precision training**: use Keras mixed-precision API, but pay attention to gradients (cancellations in Bfloat16, overflows (inf) in FP16)
  - Availability of **16-bit floating-point allows on-device training** (re-training)

- **FP32 → Integer: Quantization**
  - Post-training: Quantization via Calibration Dataset
  - Re-training with Quantization
  - Both can be done using the Ascend Model Compression Toolkit (AMCT) [5]

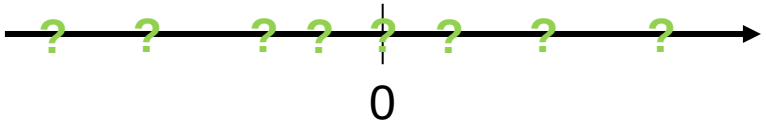
[5] CANN V199R929C10 Development Auxiliary Tool Guide. Huawei.
https://support.huawei.com/enterprise/en/doc/EDOC1100164848/aa7900c5/ascend-model-compression-toolkit-instructions-tensorflow

Weight Quantization

Weights in NNs are centered around zero

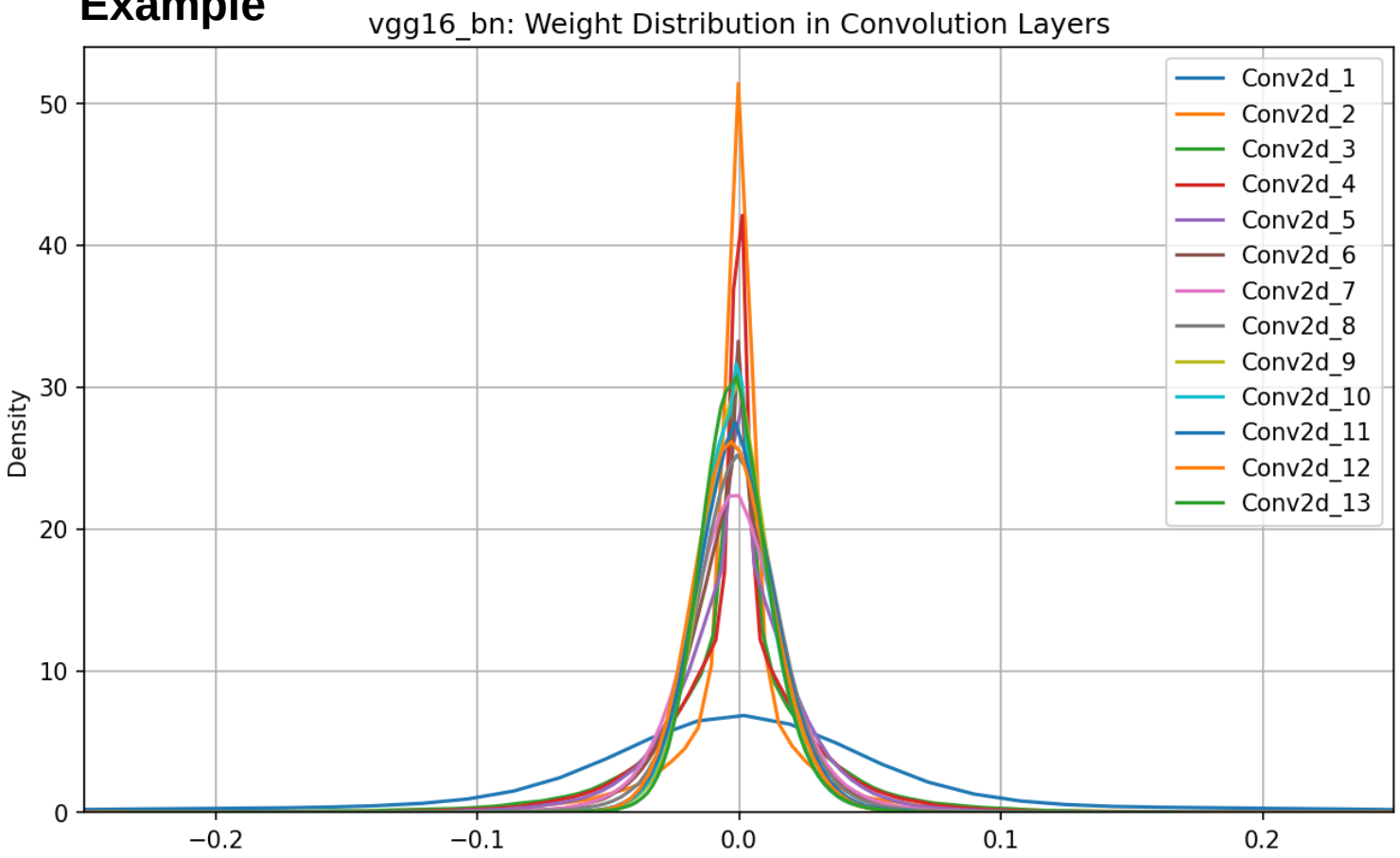


Quantization using only a subset



n-bit quantization → 2^n points

Example



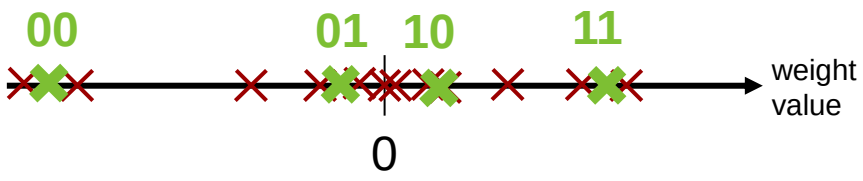
Quantiziation: Codebook Approach

= weight sharing (weights in same cluster share value)

- For a given FP32 weight matrix $\boldsymbol{W} \in \mathbb{R}^{n \times m}$ perform **k-means clustering**, with $l = 2^b$ centroids, chosen that

$$\sum_{i=1}^{m} \sum_{j=1}^{n} \sum_{k=1}^{l} \left\| w_{ij} - c_k \right\|_2^2 \longrightarrow \min \quad .$$

  - Select the closest centroid $c_{k'}$ when encoding w_{ij}. → Encode k' using b bits. [6]

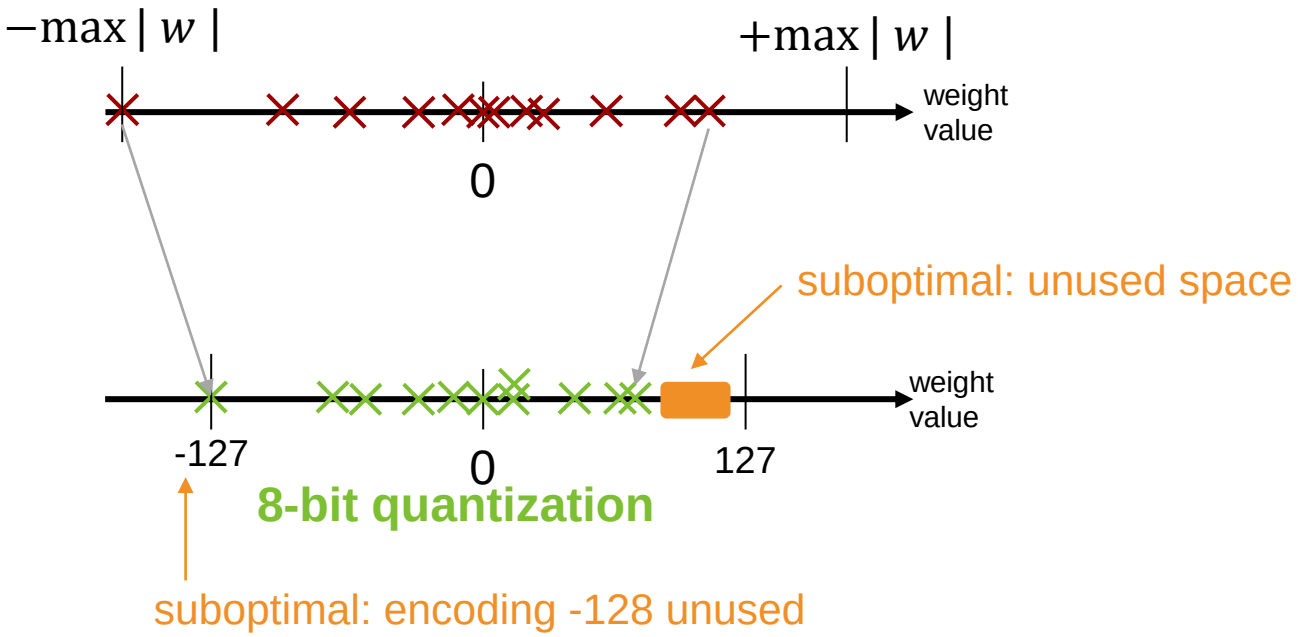


  - Store centroids $c_1, \ldots, c_{2^b}$ in codebook of length $2^b \times$ FP32.
  - Decode k': Lookup $c_{k'}$ in codebook.
  - Han et al. [7] proposes retraining the network to fine-tune the centroids. Backpropagation updates centroids (w/o reassigning weights to cluster centroids)

[6] Gong et al. Compressing Deep updates Networks using Vector Quantization. arXiv 1412.6115. 2014.
[7] Han et al. Deep Compression: Compressing DNNs with Pruning, Trained Quantization, and Huffman Coding. ICLR 2015.

Linear Quantization

- **Symmetric Quantization** $Q(w) = \text{Round}\left(\frac{2^{b-1}-1}{\max|w|}w\right)$



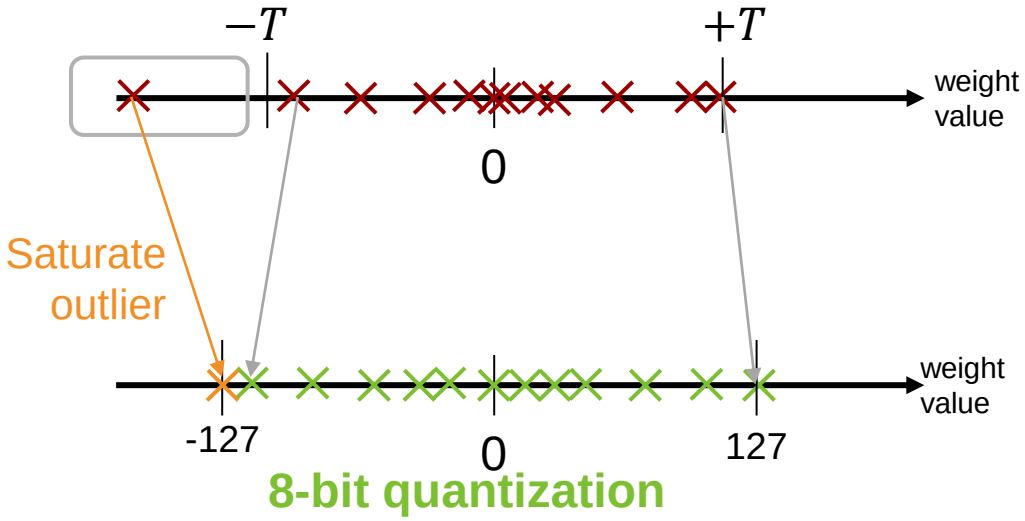
Encoding is inefficient:
Outliers increase distance between codes
→ unused space but large accuracy loss

max(..) aggregate function is very sensitive to outliers.

Robust Linear Quantization

- **Saturate values $|w| > T$**

$$Q(w) = \max\left[\min\left(\text{Round}\left(\frac{2^{b-1}-1}{T}w\right), 2^{b-1}-1\right), -2^{b-1}+1\right]$$



How do we choose Threshold T?

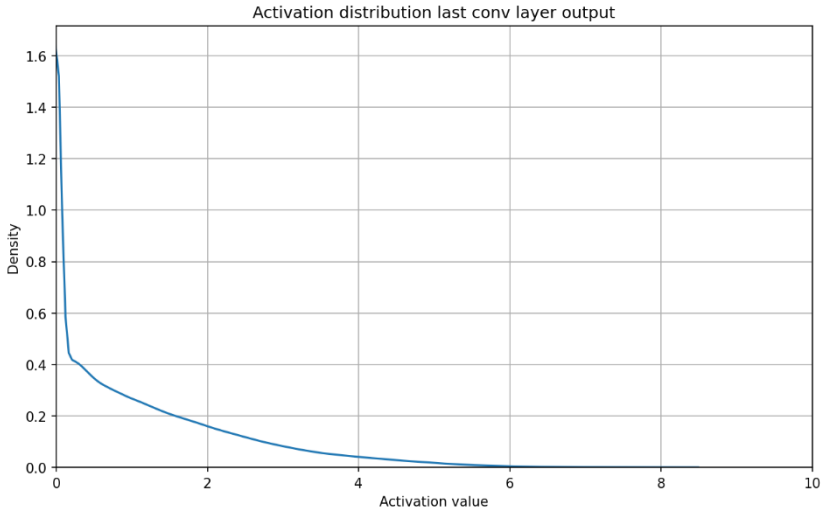
Quantization: Information Theory Perspective

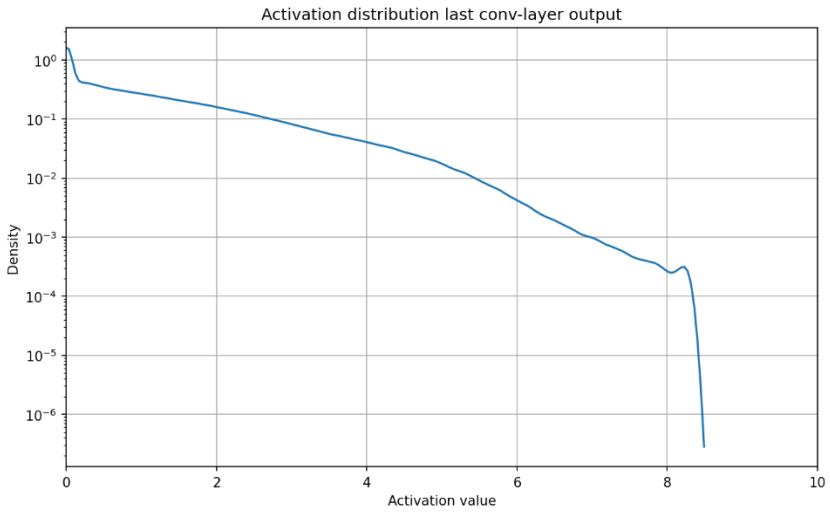
- Quantized **`INTx`** model should contain the **same information** as the FP32 model.
- Find quantization that **minimizes the information loss**.
- **Approach**: treat trained model applied on a given dataset as from some distribution $P(x)$, quantized values from a different distribution $Q(x)$.

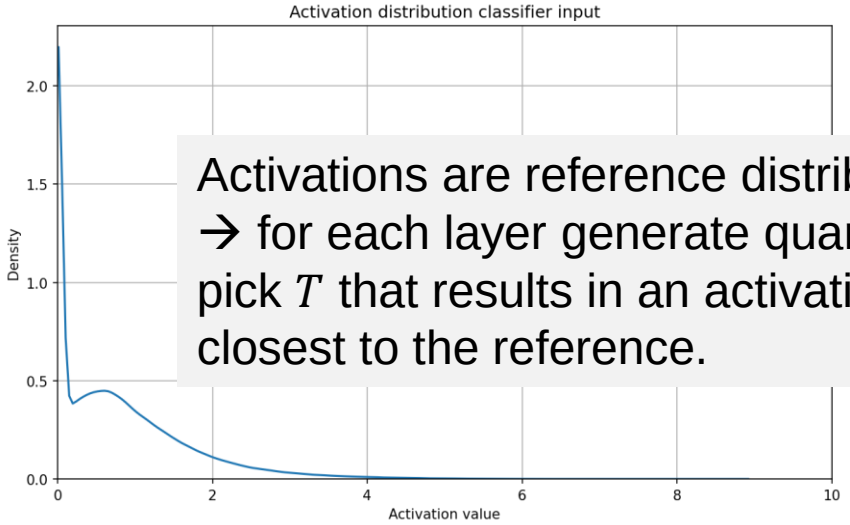
- Use Entropy of empirical distribution as a measure of information.
- Find quantization $Q(x)$ that **minimizes relative entropy** (= information loss)
- Relative entropy is the **Kullback-Leibler (KL) Divergence** between $P(x)$ and $Q(x)$:
  - $D_{KL} = -\sum_x P(x) \log\left(\frac{P(x)}{Q(x)}\right) \longrightarrow \min$.
- Pick Threshold T that KL Divergence is minimized by **iteratively** sweeping T while gathering statistics of activations on **calibration dataset** for each layer. [8]

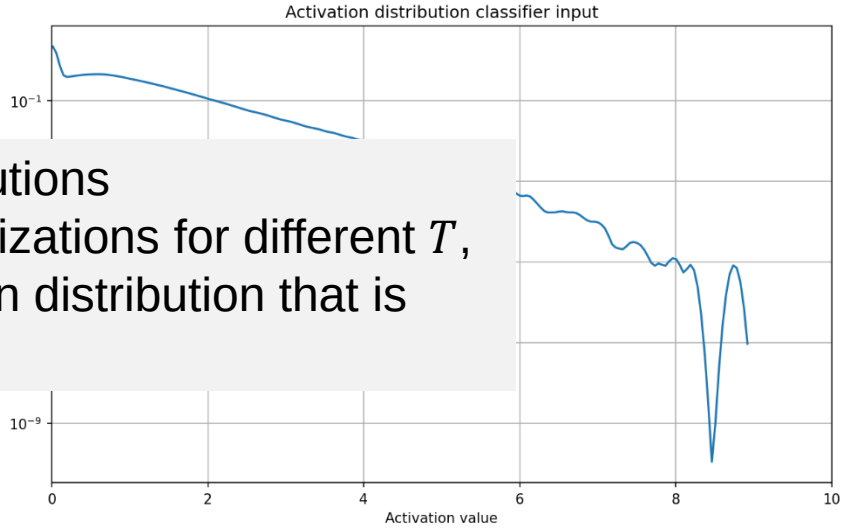
[8] Migacz. 8-bit Inference with TensorRT. In GTC, 2017.

Example: Distribution of Activations

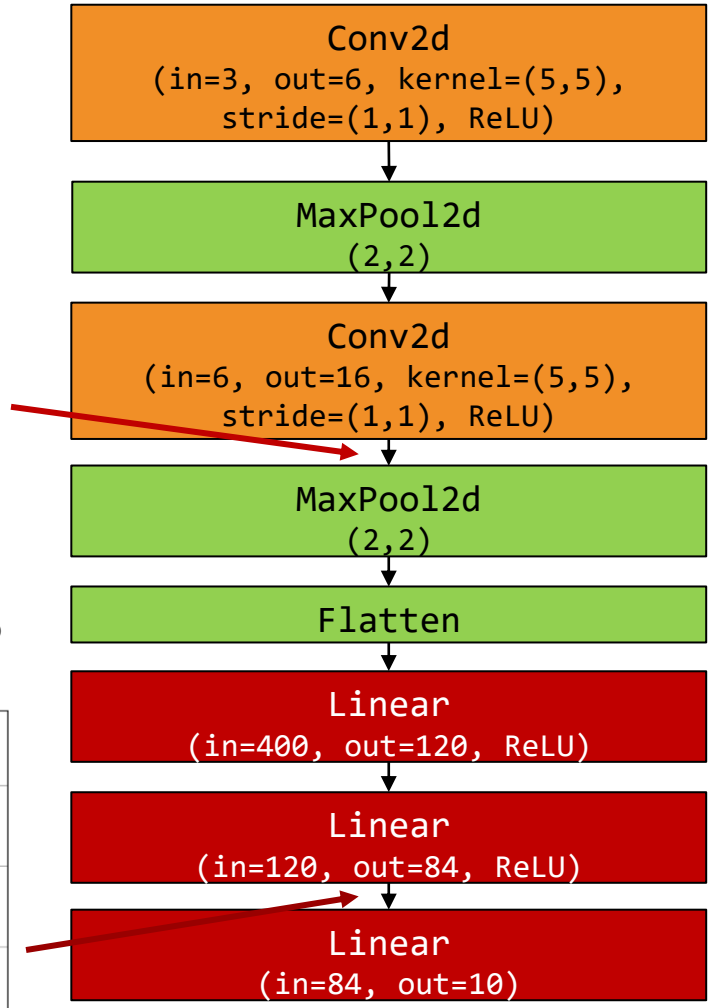




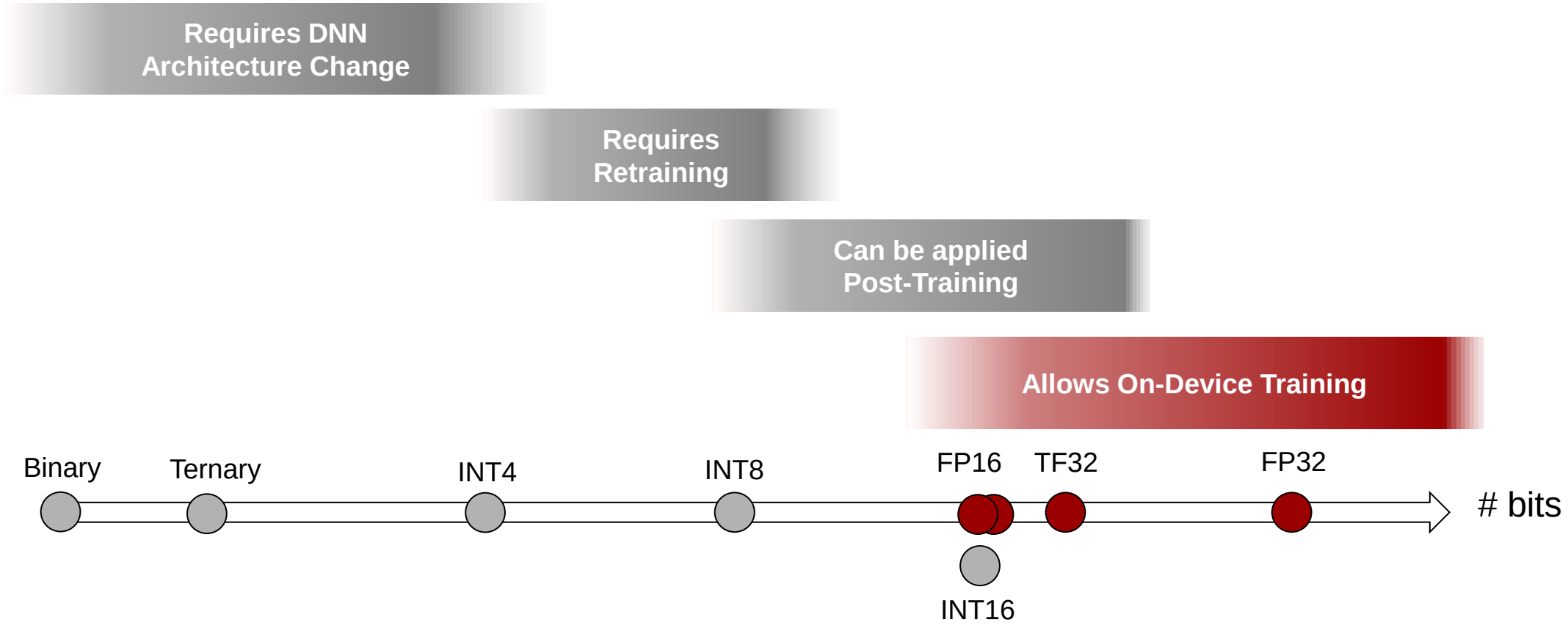




Activations are reference distributions
→ for each layer generate quantizations for different T, pick T that results in an activation distribution that is closest to the reference.



How-low can one go?



Binary Neural Networks and XNOR Networks

weights in convolution kernel (filter)

- **Binary Weight Networks: Weights $\boldsymbol{w} \in \{-1, +1\}$, real-value tensors**
  - Convolution without multiplication: w =+1 → addition, $w = -1$ → subtraction
- **XNOR-Networks: Binary Weights and Tensors**
  - Mapping Binary value -1 → **0**, value +1 → **1**
  - Multiplication: XNOR (element-wise multiplication of vector → bitwise XNOR)

x_i	y_i	$x_i * y_i$
-1	-1	+1
-1	+1	-1
+1	-1	-1
+1	+1	+1

⟺

a_i	b_i	a_i XNOR b_i
0	0	1
0	1	0
1	0	0
1	1	1

(popcount: number of 1s in bitvector)

  - Dot-product $\langle \boldsymbol{x}, \boldsymbol{y} \rangle$ of vectors $\boldsymbol{x}, \boldsymbol{y}$ $\Leftrightarrow \mathrm{popcount}(\boldsymbol{a} \text{ xnor } \boldsymbol{b})$

[9] Rastegari et al. XNOR-Net: ImageNet Classification Using Binary Convolution Neural Networks. In ECCV 2016.
[10] Rastegari et al. Enabling AI at the Edge with XNOR-Networks. In Comm of the ACM. Vol 63, No. 12, pp 83-90, 2020.

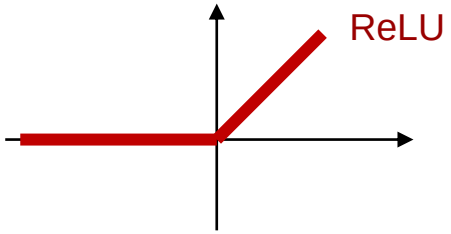
Binary Neural Networks and XNOR Networks

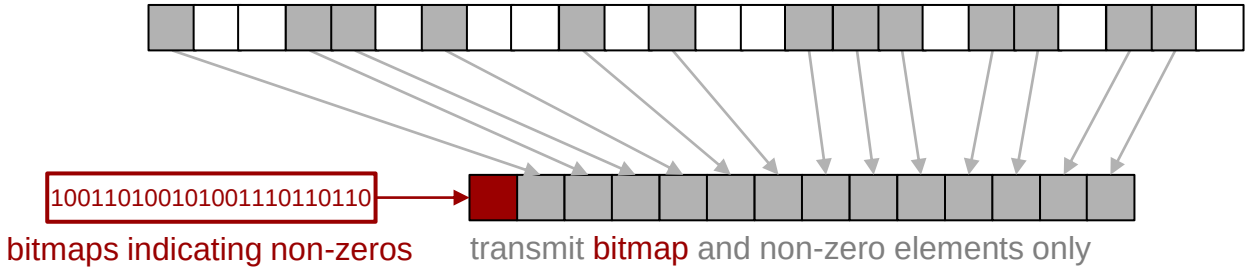
- Training is challenging, binarization (`sign` is non differentiable everywhere and where it is derivative is zero) use $\max(-1, \min(1, x))$ or identity function for backprop
- Sometimes a distillation approach can be used with a teacher in full-precision and binary student network.
- Batch Norm cannot be fused into weights (as a linear transformation as you can in higher precision)
- Binary neural networks almost always require architecture changes to obtain similar performance as full precision networks, e.g., adding more layers.
  - → diminishing returns when accuracy is pushed to high

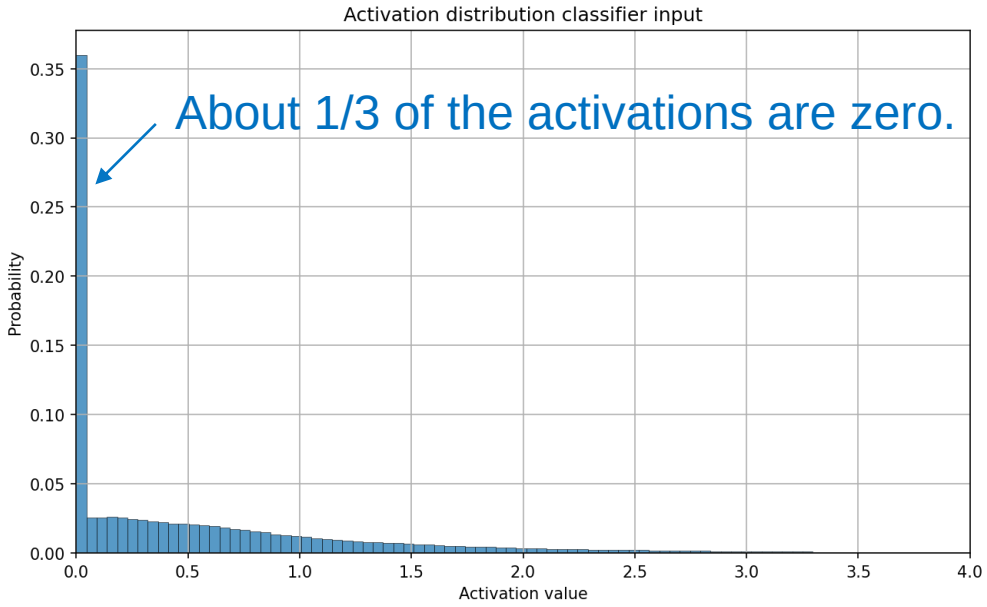
Compression: Sparsity of Activations

- **ReLU is extremely popular activation function**
  - Non-vanishing gradient > 0
- **Output of ReLU is 0 for for ½ of the inputs → sparse output**
- **Do you need to process zeros?**
  - Zero-bit use zero-bit line to “disable” multiplier → saves energy.
- **Do you need to transfer zeros?**

  → Zero-value Compression [12]



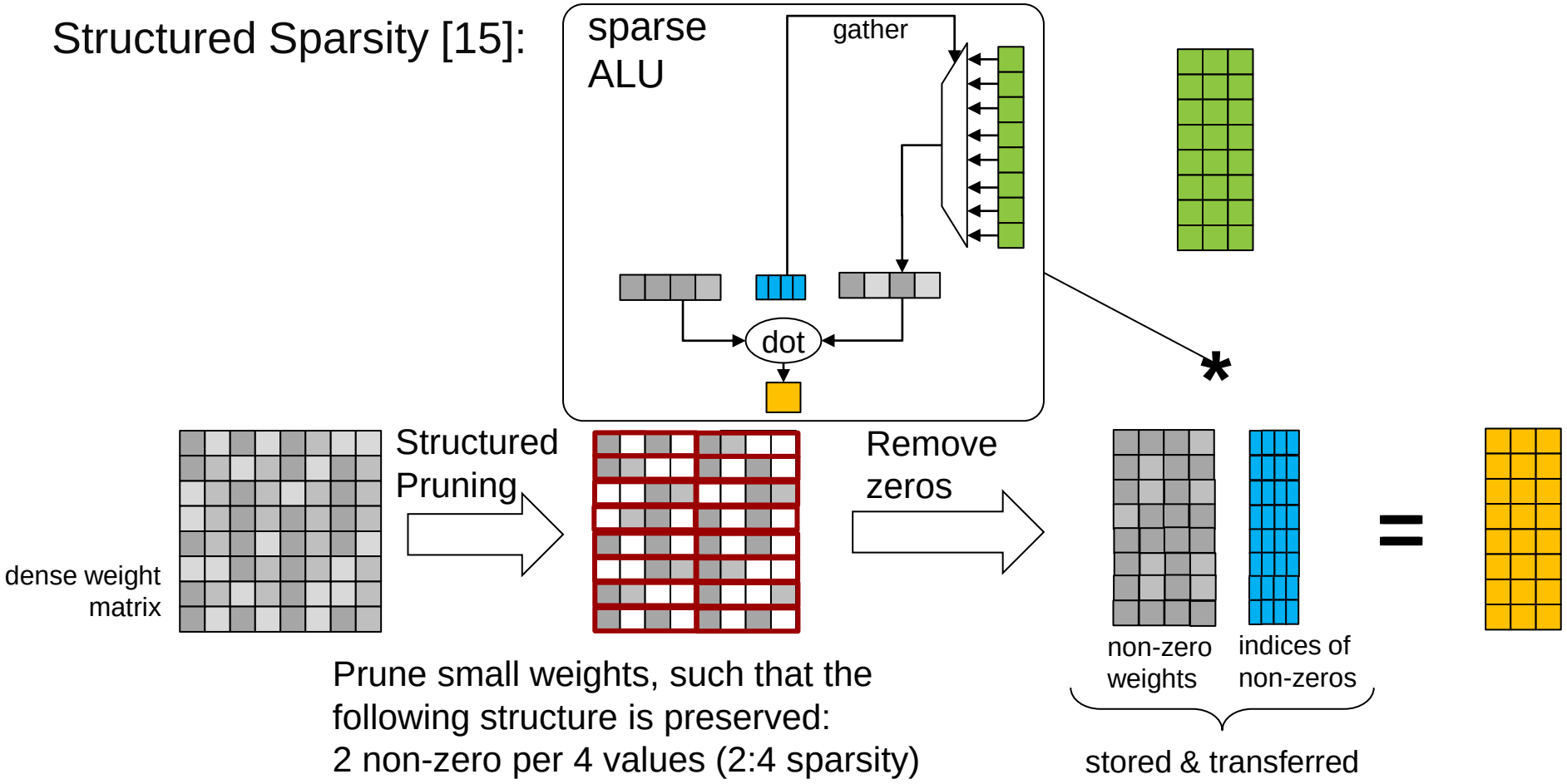




[12] Rhu et al. Compressing DMA Engine: Leveraging Activation Sparsity for Training Deep Neural Networks. In HPCA 2018.

Compression: Sparsity of Models

- Weight pruning for regularization (at the cost of decreased noise resiliency [14]).
- Structured Sparsity [15]:



[14] Liebenwein: The Effect of Pruning Neural Networks beyond Test Accuracy. In MLSys 2021.
[15] NVIDIA. A100 Whitepaper. 2020.

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Outline

1) Quantization

2) **Huawei DaVinici Core and Ascend SoC**

Deep Learning Hardware

Huawei DaVinci Core and Ascend SoC

based on Presentation by Heiko Joerg Schick (Chief Architect Advanced Computing, HiSilicon)

https://www.schihei.eu/blog/archives/03-2021

Technology Scalable Across Device Classes

	Device				Edge		Cloud
	Earphone	**Always-on**	**Smartphone**	**Laptop**	**IPC**	**Edge Server**	**Data Centre**
Compute	20 MOPS	100 GOPS	1 - 10 TOPS	10 - 20 TOPS	10 - 20 TOPS	10 - 100 TOPS	200+ TOPS
Power budget	1 mW	10 mW	1 - 2 W	3 - 10 W	3 - 10 W	10 - 100 W	200+ W
Model size	10 KB	100 KB	10 MB	10 - 100 MB	10 -100 MB	100+ MB	300+ MB
Latency	< 10 ms	~10 ms	10 - 100 ms	10 - 500 ms	10 - 500 ms	ms ~ s	ms ~ s
Inference?	Y	Y	Y	Y	Y	Y	Y
Training?	N	N	Y	Y	Y	Y	Y
SoC scale	Nano	Tiny	Lite	Mini Ascend 310	Mini Ascend 310	Multi-Mini Ascend 310	Max Ascend 910

Two System-on-Chip (SoC)

Inference

Ascend 310

AI SoC with ultimate efficiency

- Half precision (FP16): 8 TFLOPS
- Integer precision (INT8): 16 TOPS
- 16-channel full-HD video decoder: H.264/265
- 1-channel full-HD video encoder: H.264/265
- Max. power consumption: 8 W
- 12nm

Training

Ascend 910

Highest compute density on a single chip

Atlas Products with Ascend Accelerators

52 mm x 38 mm x 10 mm

Atlas 300

AI Accelerator Card

- 64 TOPS of INT8 @ 67 W
- 32 GB memory
- 64-channel HD video real-time analytics
- Standard half-height half-length PCIe card form factor, applicable to general-purpose servers

reddot award 2019 winner

Atlas 200

AI Accelerator Module

- 16 TOPS of INT8
- 16-channel HD video real-time analytics, JPEG decoding
- 4 GB/8 GB memory, PCIe 3.0 x4 interface
- Operating temperature: -25°C to +80°C

Atlas 500

AI Edge Stations

- 16 TOPS of INT8
- 25–40 W
- Wi-Fi & LTE
- 16-channel HD video real-time analytics
- Fanless design, -40°C to +70°C environments

reddot award 2019 winner

Atlas 800

Deep Learning System

- Plug-and-play installation
- Ultimate Performance
- Integrated Management

Ascend 310

AI SoC with ultimate efficiency

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- Max. power consumption: 8 W
- 12nm

Ascend 910

Highest compute density on a single chip

- Half precision (FP16): 256 TFLOPS
- Integer precision (INT8): 512 TOPS
- 128-channel full-HD video decoder: H.264/265
- Max. power consumption: 350 W
- 7nm

Atlas 900 AI Cluster

The pinnacle of computing power

- Thousands of Ascend 910 AI processors
- High-speed interconnection
- Delivers up to 256 to 1024 PetaFLOPS at FP16
- Can complete model training based on ResNet-50 within 59.8 seconds
- 15% faster than the second-ranking product
- Faster AI model training with images and speech

Atlas 200 DK

Quickly build development environments in 30 minutes

- 16 TOPS of INT8 @ 24 W
- 1 USB type-C, 2 camera interfaces, 1 GE port, 1 SD card slot
- 4 GB/8 GB memory

Kunpeng 920

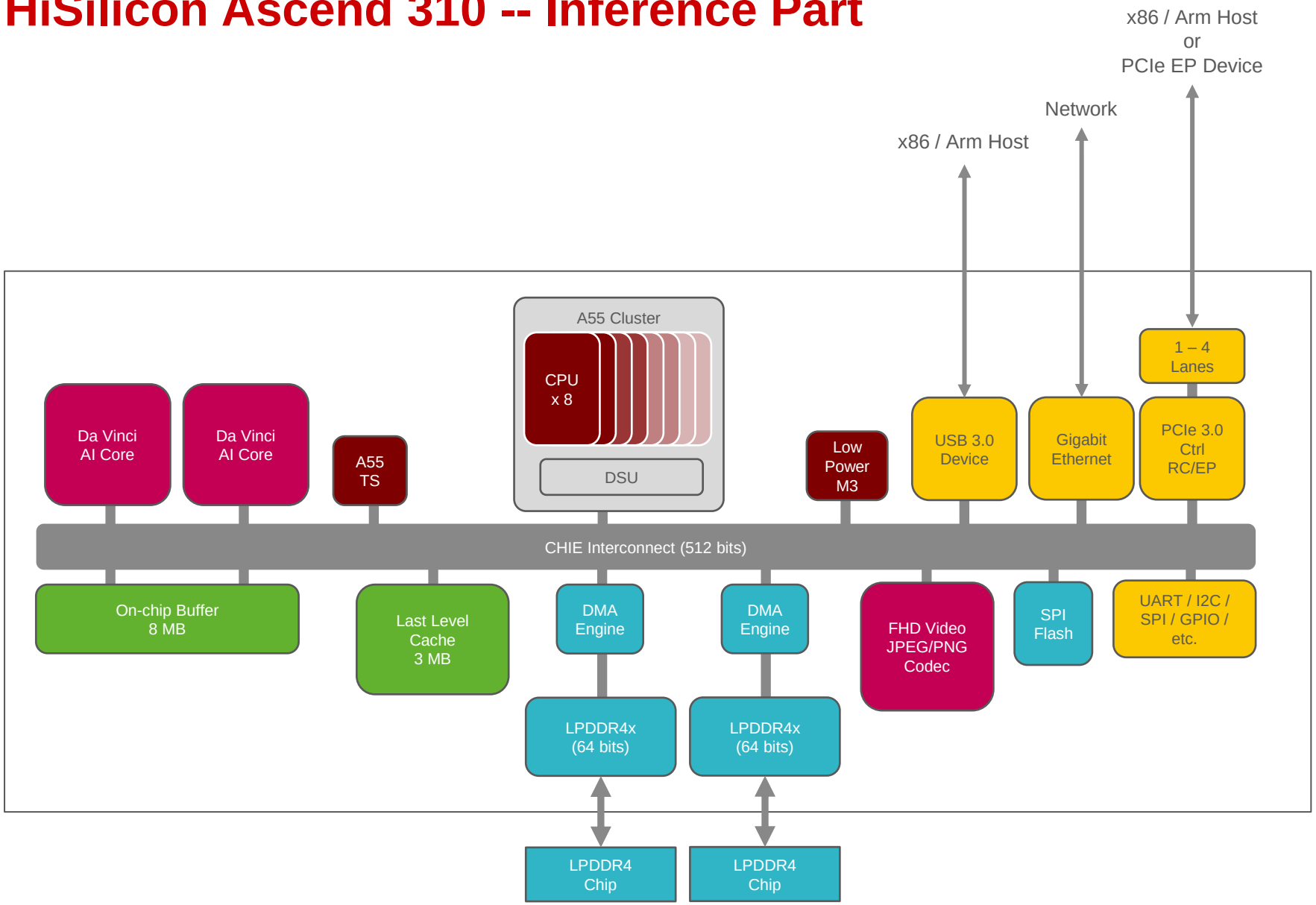
The industry's highest-performance ARM-based server CPU

- ARM v8.2-architecture
- up to 64 cores, 2.6 GHz
- 8 DDR4 memory channels
- PCIe 4.0 and CCIX
- Integrated 100GE LOM and encryption and compression engines
- Supports 2- or 4-socket interconnects

Storage-intensive — **Computing-intensive**

5290	5280	2280	1280	2480	X6000
4U 72-drive storage model	4U 40-drive storage model	2U 2S balanced model	1U 2S high-density model	2U 4S high-performance model	2U 4-node high-density model

HiSilicon Ascend 310 -- Inference Part



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HiSilicon Ascend 910 -- Training Part

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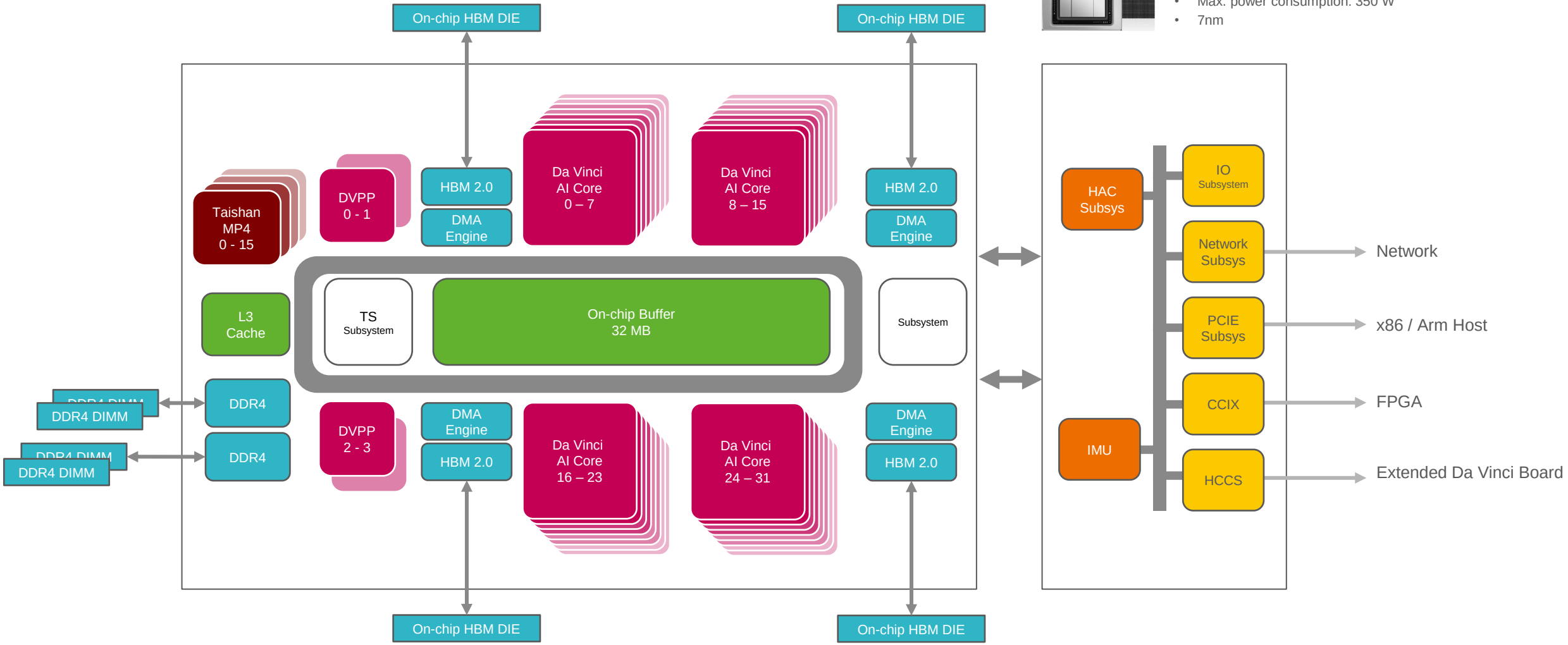
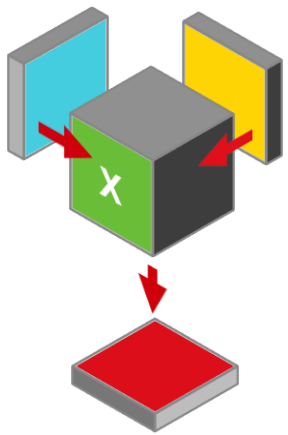
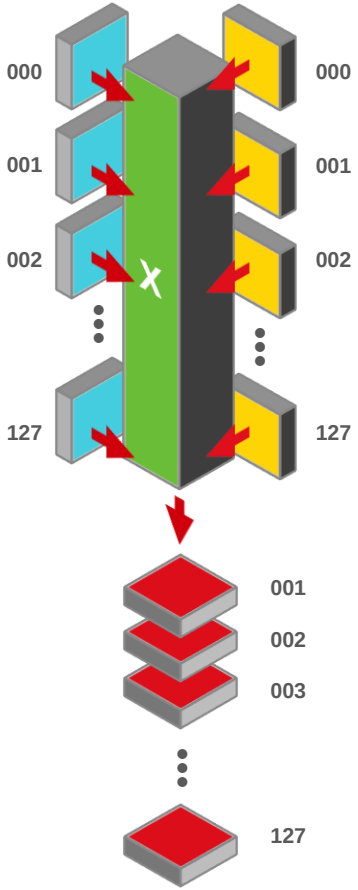


Building Blocks for Deep Learning Accelerators

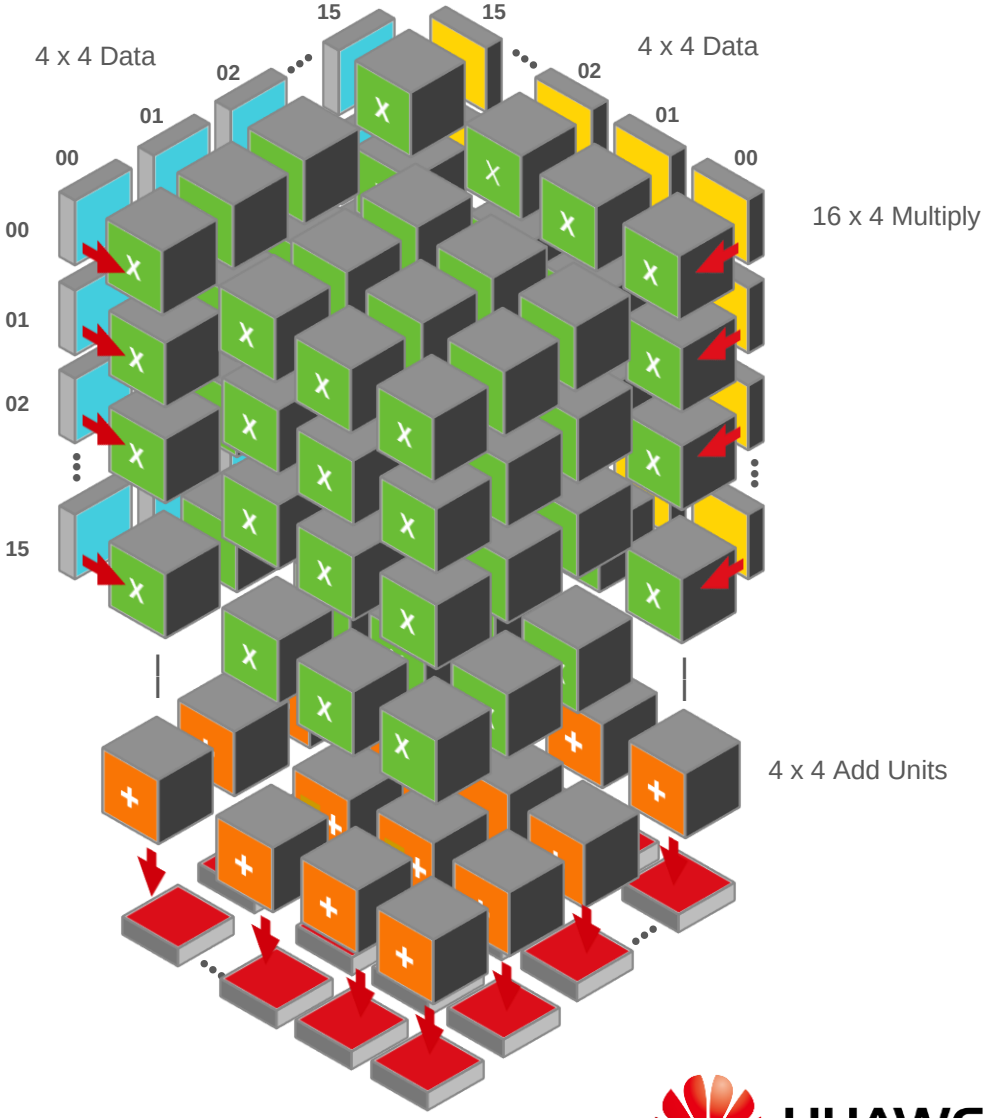
Scalar Unit
Full flexibility in computation



Vector Unit
Rich and efficient operations



Cube Unit
High intensity computation



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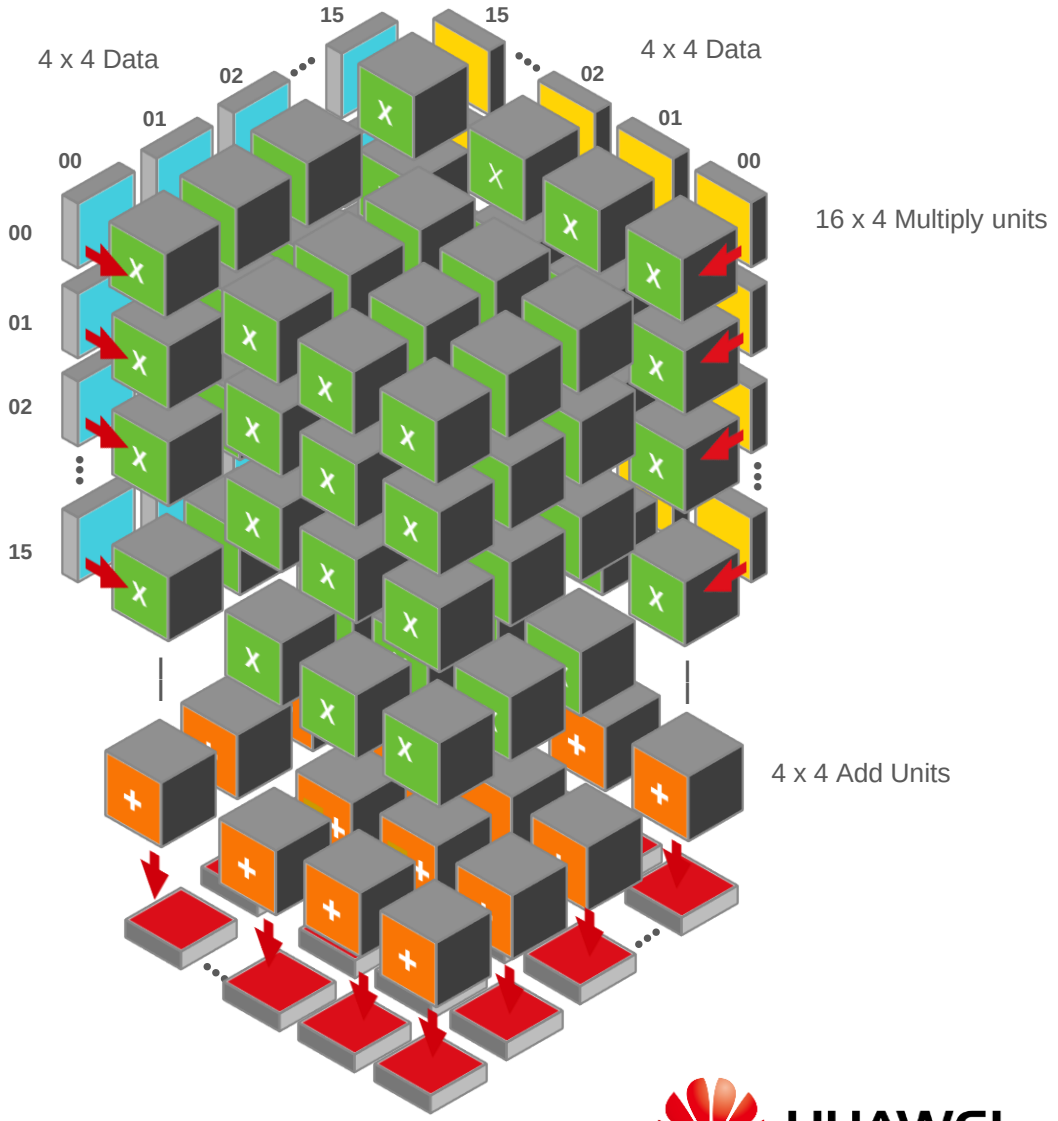
Building Blocks for Deep Learning Accelerators

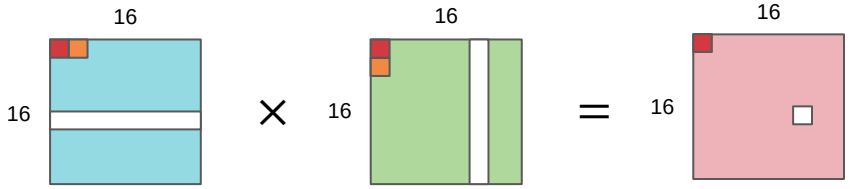
N	N^2	N^3
1	1	1
2	4	8
4	16	128
8	64	512
16	**256**	**4,096**
32	1,024	32,768
64	4,096	262,144

	GPU + Tensor core	AI core + SRAM
Area (normalized to 12 nm)	5.2 mm^2	13.2 mm^2
Compute power	1.7 TFLOPS (FP16)	8 TFLOPS (FP16)

Cube Unit
High intensity computation



Advantages of Special Compute Units




```
float a[16][16], b[16][16], c[16][16];
```

Scalar:

```
for (int i=0; i<16; i++) {
    for (int j=0; j<16; j++) {
        for (int k=0; k<16; k++) {
            c[i][j] += a[i][k] * b[k][j];
        }
    }
}
```

$Cycles = 16 * 16 * 16 * 2 = 8192$
$Data\ per\ cycle = Rd\ 3;\ \ Wr\ 1$

Vector:

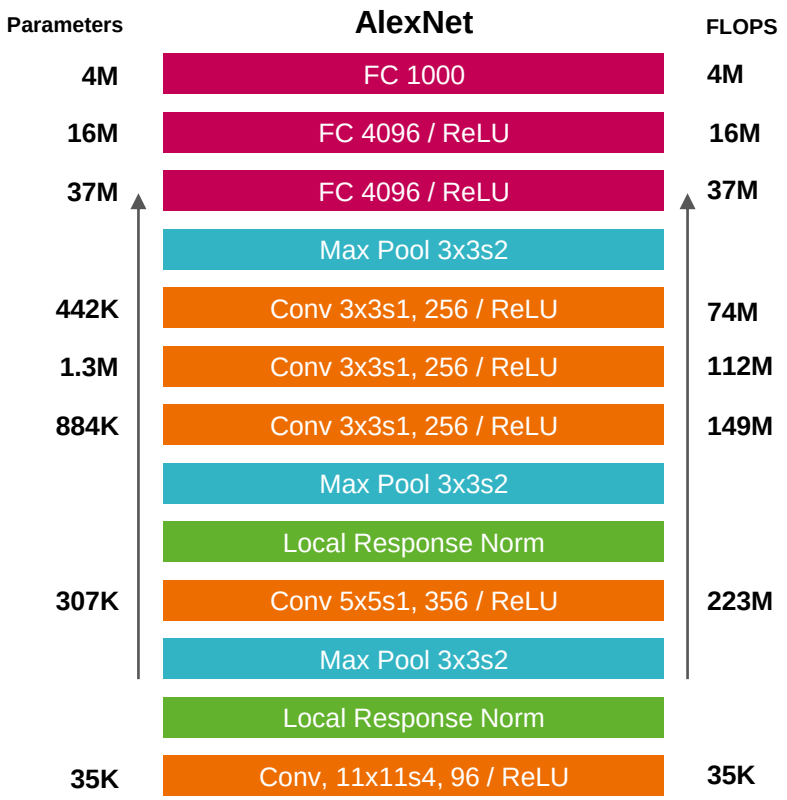
```
for (int i=0; i<16; i++) {
    for (int j=0; j<16; j++) {
        c[i][j] = a[i][:] *+ b[:][j];
    }
}
```

$Cycles = 16 * 16 = 256$
$Data\ per\ cycle = Rd\ 2 * 16;\ \ Wr\ 16$

Cube:

```
c[:][:] = a[:][:] X b[:][:]
```

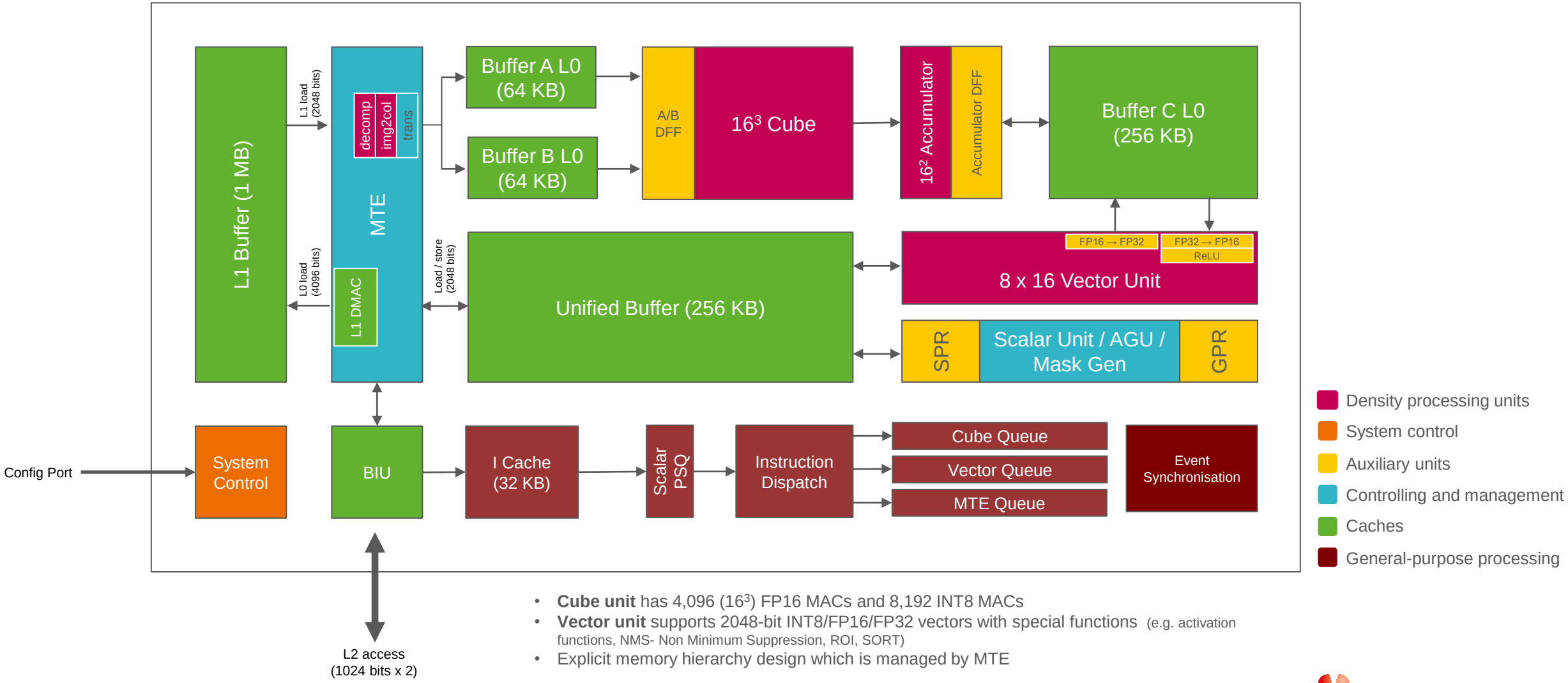
$Cycles = 1$
$Data\ per\ cycle = Rd\ 2 * 16 * 16;\ \ Wr\ 16*16$



Number of parameters and floating point operations per second (FLOPS) for each layer of the AlexNet artificial intelligence model.

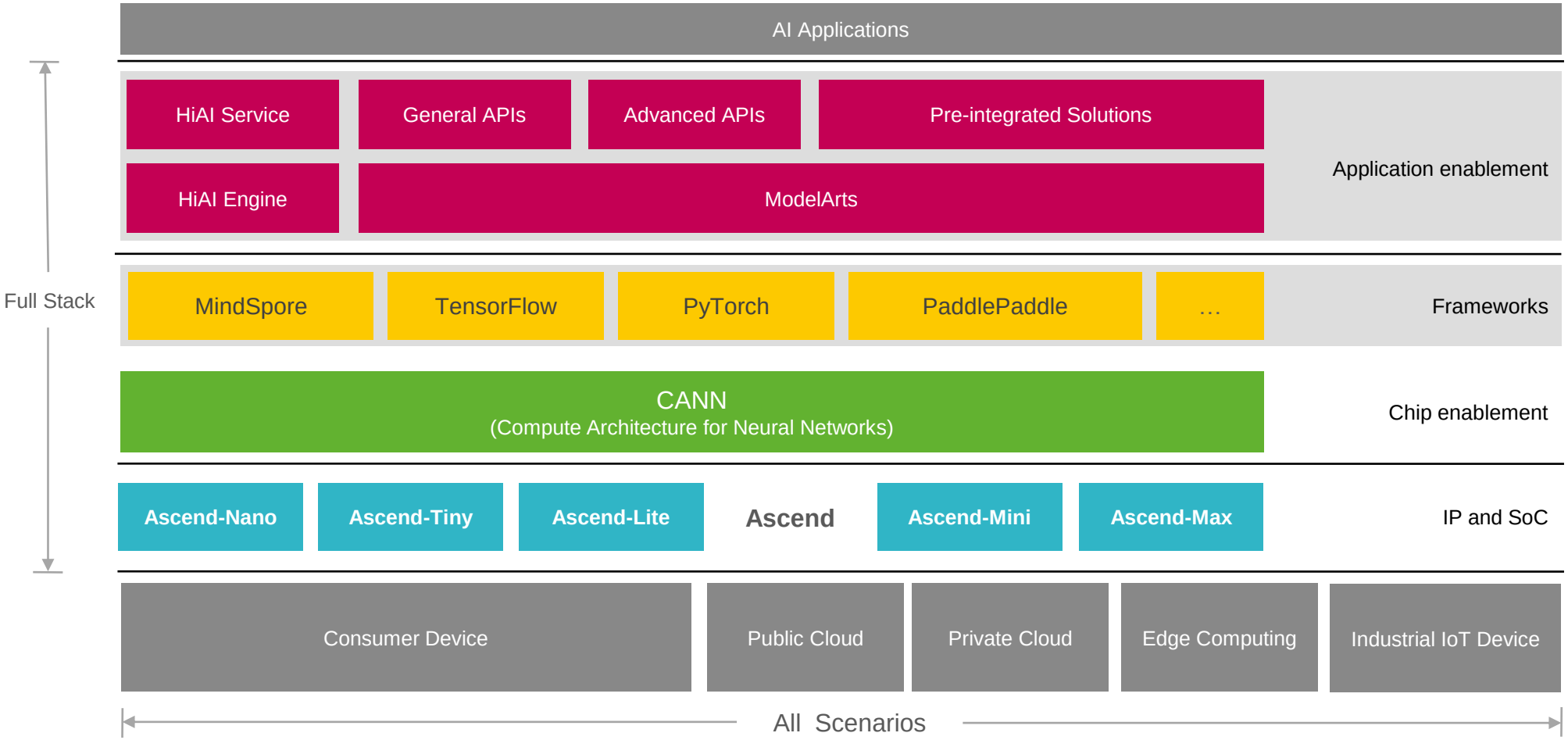
99% of the computations are matrix-matrix multiplications	Typical CNN networks		
	AlexNet	**VGG16**	**Inception-v3**
Model memory (MB)	> 200	> 500	90-100
Parameter count (Million)	60	138	23.2
Computation amount (Million)	720	15300	5000

DaVinici Core Architecture



- **Cube unit** has 4,096 (16^3) FP16 MACs and 8,192 INT8 MACs
- **Vector unit** supports 2048-bit INT8/FP16/FP32 vectors with special functions (e.g. activation functions, NMS- Non Minimum Suppression, ROI, SORT)
- Explicit memory hierarchy design which is managed by MTE

Ascend AI Software Stack



Application enablement:
Full-pipeline services (ModelArts), hierarchical APIs, and pre-integrated solutions

MindSpore:
Unified training and inference framework for device, edge, and cloud (both standalone and cooperative)

CANN:
SoC operator library and highly automated development toolkit

Ascend:
AI IP and SoC series based on unified scalable architecture

MindSpore Deep Learning Framework

https://www.mindspore.cn/en

MindSpore

```
import mindspore.dataset as ds
from mindspore import context, nn, ops
from mindspore.train import Model
from mindspore.train.callback import LossMonitor

class Net(nn.Cell):
    def __init__(self, input_dim, hidden1_dim, hidden2_dim):
        super().__init__()
        self.fc1 = nn.Dense(in_channels=input_dim,
            out_channels=hidden1_dim, activation='relu')
        self.fc2 = nn.Dense(in_channels=hidden1_dim,
            out_channels=hidden2_dim, activation='relu')
        self.fc3 = nn.Dense(in_channels=hidden2_dim,
            out_channels=1)

    def construct(self, input_x):
        h1 = self.fc1(input_x)
        h2 = self.fc2(h1)
        return self.fc3(h2)


class Loss(nn.Cell):
    def __init__(self):
        super().__init__()
        self.loss_fn = ops.SigmoidCrossEntropyWithLogits()
        self.mean_fn = ops.ReduceMean()

    def construct(self, logits, labels):
        return self.mean_fn(self.loss_fn(logits, labels))
```

Graph Mode: static compute graph (~ non-eager execution)

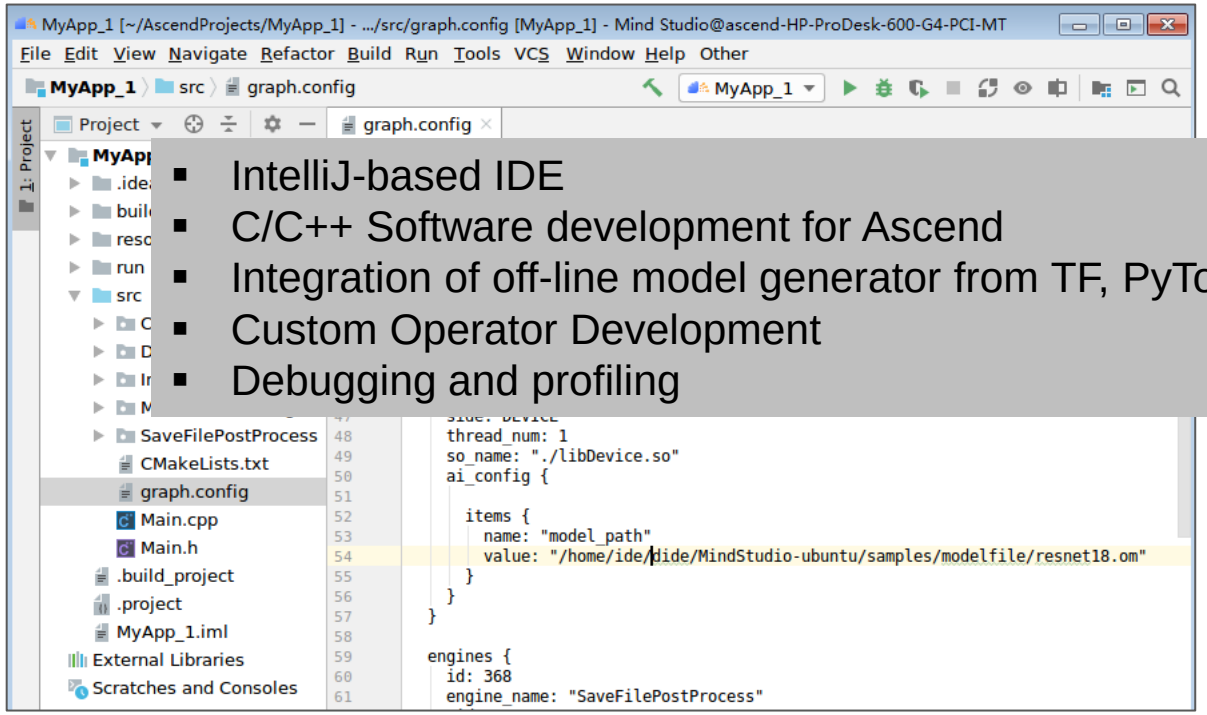
Execution on Ascend (other options ‘GPU’ or ‘CPU')

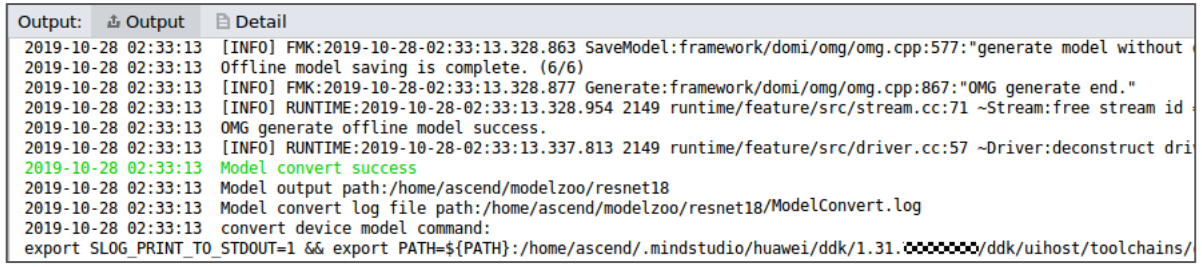
```
...
context.set_context(
    mode=context.GRAPH_MODE,
    device_target='Ascend')
dataset = create_dataset(num_rows=10000, input_dim=5)
dataset = dataset.batch(batch_size=128)

net = Net(input_dim=5, hidden1_dim=100, hidden2_dim=20)
loss_obj = Loss()
net_opt = nn.Adam(net.trainable_params())
model = Model(network=net, loss_fn=loss_obj, optimizer=net_opt)

model.train(epoch=400, train_dataset=dataset,
    callbacks=[LossMonitor(per_print_times=100)],
    dataset_sink_mode=False)
```

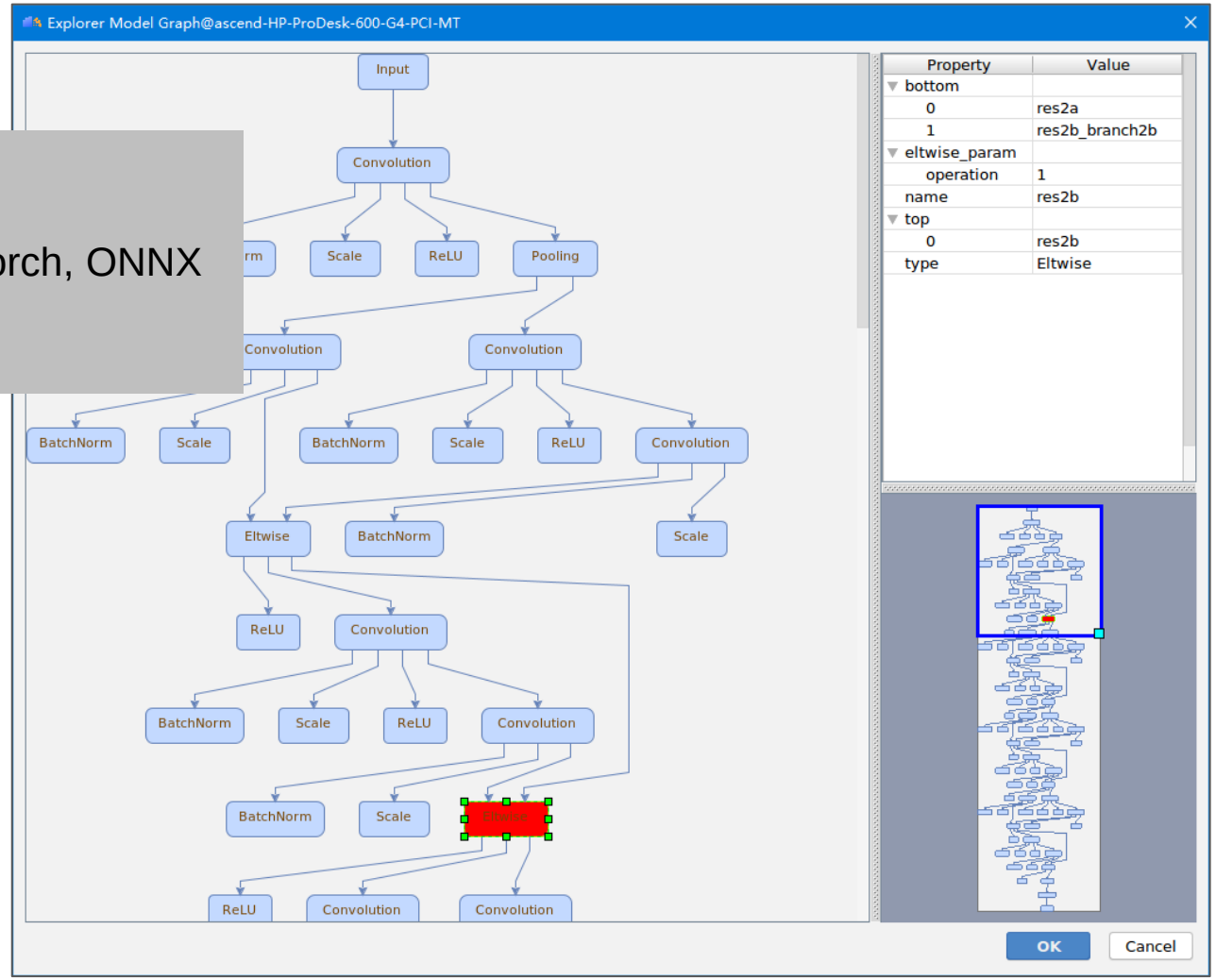
https://www.hiascend.com/software/mindstudio

MindStudio IDE

- IntelliJ-based IDE
- C/C++ Software development for Ascend
- Integration of off-line model generator from TF, PyTorch, ONNX
- Custom Operator Development
- Debugging and profiling







HUAWEI

https://developer.huawei.com/consumer/en/hiai

HiAI Engine and HiAI Foundation (HMS)

- **HMS (Huawei Mobile Services) set of services for the mobile platform**
- **AI related components:**

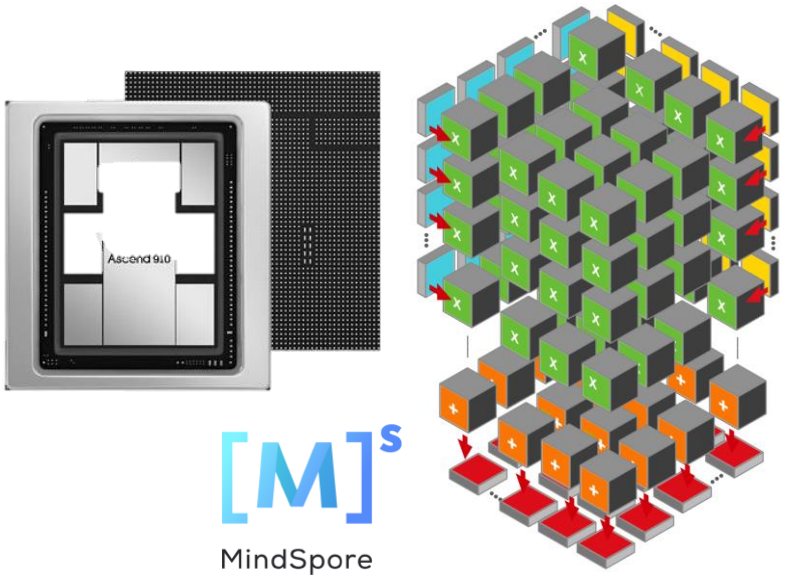
Cloud-enabled Services: Abilities in the cloud
AI Voice, AI Lens, and distribution.

Use existing AI building blocks for mobile apps
Domains: Image Recognition, Pose Estimation,
Text Recognition

Lowest level: Convert existing models for
HW acceleration and heterogenous scheduling
Android Studio + NDK native libraries with C++ (JNI)

Cloud	**HUAWEI HiAI Service**
Device	**HUAWEI HiAI Engine**
Chip	**HUAWEI HiAI Foundation**

Conclusions



- **Deep Learning is an important workload**
  → Building custom Hardware makes sense

- **Efficient Inference is critical for application**
  - Energy efficiency
  - Low costs
  - Low latency

- **Up to now: DL Accelerators = Matrix-Multiplication Accelerator**
- **Future: Accelerator for Sparse Data and DNN, near-data and in-memory processing**
- **Software (Framework support) and Tooling equally important**

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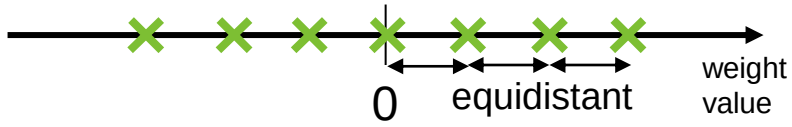
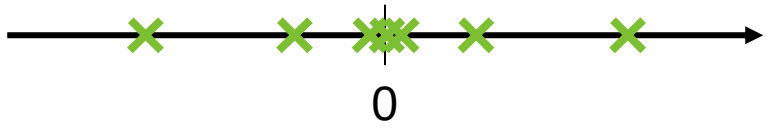
Logarithmic Number Systems

- **Weights are centered around zero.**
- **Numbers around zero should have good representation.**

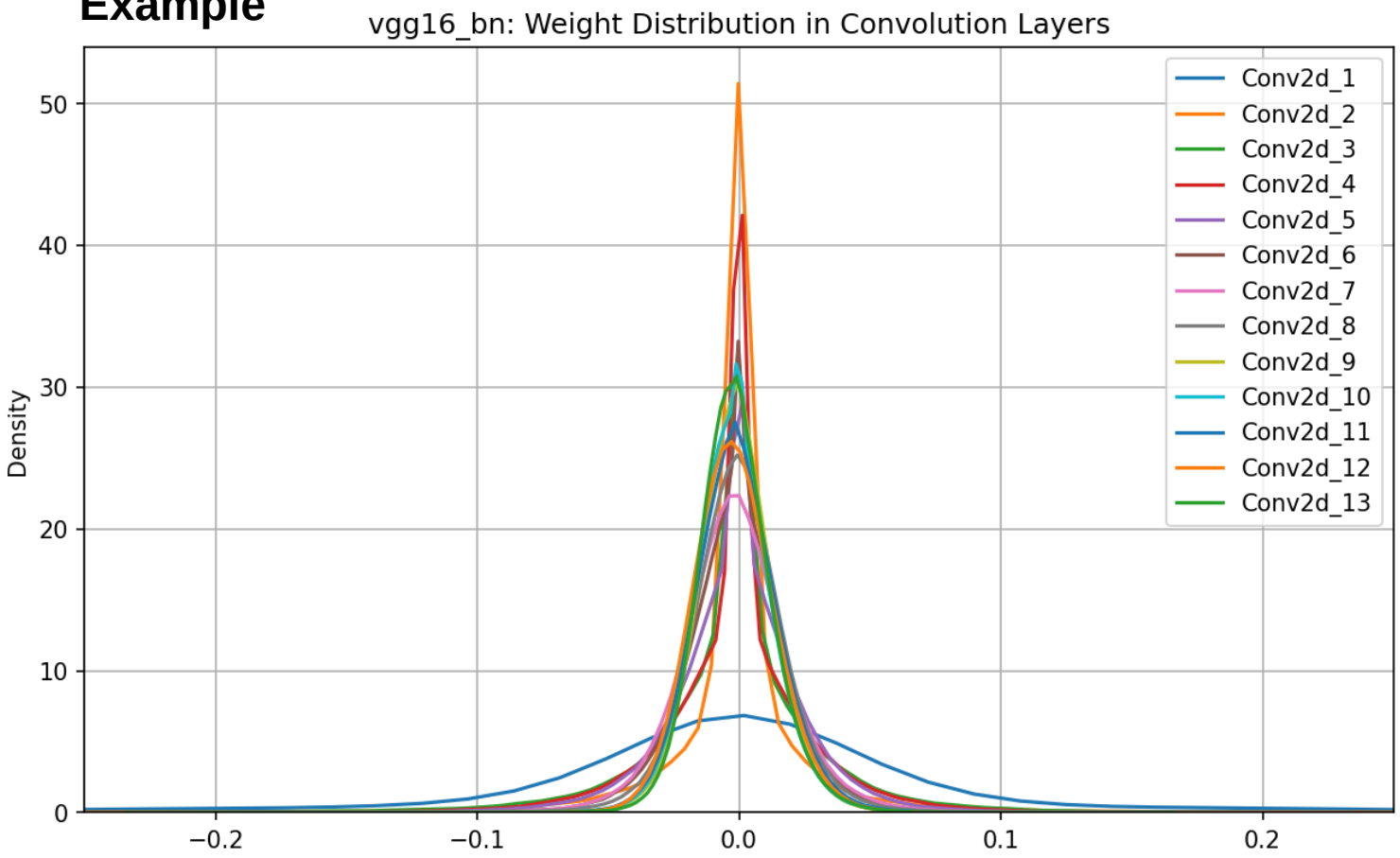
Linear Quantiziation



Logarithmic Quantiziation



Example



[11] Schwartzlander, Alexopoulos: The Sign/Logarithm Number System. In IEEE Trans. of Computers, Vol. C-24, Issue 12, 1975.

Logarithmic Number Systems

$$v = (-1)^s 2^{e.f}$$

Log4.3

1	4	3
s	e	f
0	1110	101

$= 2^{-1.375} \approx 0.385552706352\ldots$

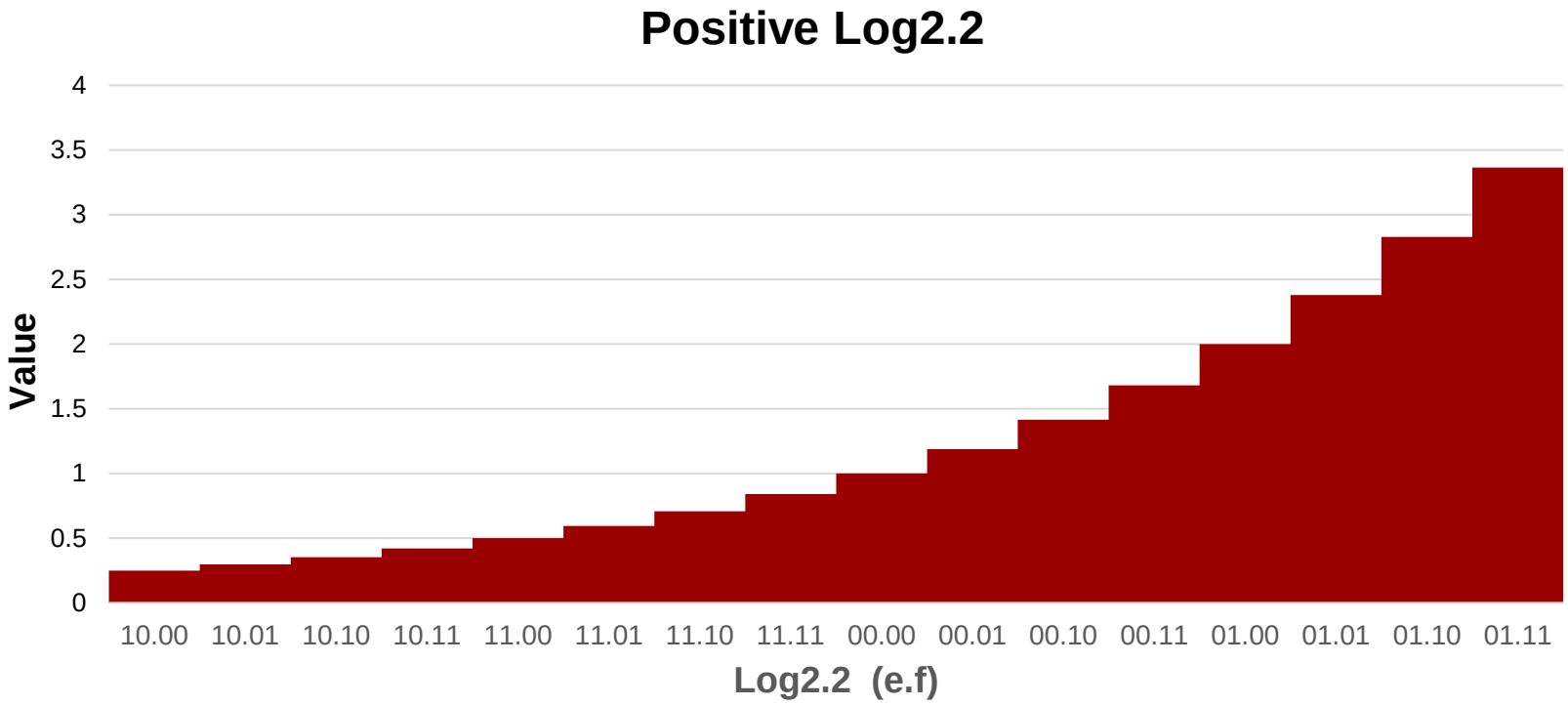
$= (1110.101)_2 = -(0001.011)_2 = -(1 + 1/4 + 1/8)$

$= -1.375$

Example

Log2.2: $v = (-1^s)2^{b_1 b_0 \,.\, b_{-1} b_{-2}}$

($e.f$ without offset)



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Logarithmic Number Systems

- **Representation:** $(-\mathbf{1})^{s} \times \mathbf{2}^{e.f}$
- **Special treatment for zero** (and special encoding)
- **Arithmetic**
  - Multiplication and Division: $\log(x \times y) = \log(x) + \log(y)$, $\log(x/y) = \log(x) - \log(y)$
  - Addition and Subtraction:

$$\log(x \pm y) = \log\left(x\left(1 \pm \frac{x}{y}\right)\right) = \log(x) + \log\left(1 + \frac{x}{y}\right) = \log(x) + \log(1 + 2^{e.f_x - e.f_y})$$

  - Addition/Subtraction requires lookup tables (LUTs) implemented as ROM for linear-to-log conversions

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Von Neumann Laboratory

Huawei Research Center, Zurich

Huawei R&D in Europe

Close to industrial and academic partners, accelerate ICT-based innovations of future life and future work.

Industry Solution Development

- Telecom
- IT
- ITS / Mobility
- Industry 4.0
- IoT
- E-/M-Health

+

Key Technical Innovation

- Cloud and Big Data
- Wireless Networks
- Optical System
- Terminals and IoTings

+

Fundamental Research

- New Theories
- New Materials
- New Components
- New Devices

Huawei R&D in Europe: Open and Joint Value Creation

1,600
industry partners

TO DEVELOP JOINT

Innovative Solutions

230 technical collaboration projects

200+ European technical experts

180+ European Academics

63M € + investment from Huawei in technical partnerships with over 100 of European top universities, research institutes and enterprises

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